Evidence update: scoped facility inventories

V124 companion to the independent V123 review draft

Two verified facility inventories improve precision while preserving the city recommendation. Madrid's registered metro subset and five public indoor pools in Tokyo replace two fully missing entries with lower bounds. The original city table, all 139 previously non-fully-missing entries, weights and utility anchors stay fixed. The resulting matrix contains 112 point values, 29 partial intervals and 19 fully missing entries.

Table 1: Two facility lower bounds improve score precision. Fixed weights and all other evidence stay fixed.

City	Old lower	New lower	Upper	Decision rank
Madrid	48.62761	49.00419	68.22419	10
Tokyo	43.80772	43.95921	61.11878	15

The average fixed-weight score-interval width falls from 16.18302 to 16.15002, a 0.204% reduction. Both fixed and equal reference weights retain their complete decision orders: Zurich remains the conservative recommendation, all sixteen cities remain possible champions, and no necessary champion is identified. Additional supply evidence narrows uncertainty without establishing observed improvement over time.

Count facilities inside the comparison boundary

Madrid's regional transport authority (CRTM) publishes a station registry with municipality codes and network-station identifiers [1]. The original 293 records include 242 tagged to Madrid city (province 28, municipality 079, independently identified by INE [2]). Two unnamed records with no route information are omitted. Consolidating the remaining 240 named records by network-station code gives 191 metro complexes. Noviciado/Plaza de España and Embajadores/Acacias are each conservatively counted once. The fifty-one outside-city records do not enter the count.

This is a 29 May 2026 registered inventory divided by the unchanged supplied 2025 population of 3,527,924. Its transport utility is $[0.09770010875, 1]$ under the existing anchor 50. Other modes and providers remain uncounted, and the registry does not establish real-time operation. It therefore supplies a conditional lower bound rather than a complete same-year city transport census.

Distinguish current closures from planned closures

Tokyo's retained subset consists of five municipal indoor swimming venues at different addresses in Nerima, one of the twenty-three special wards [3]. Each facility's original municipal page confirms a 25-metre pool and regular opening information. The current closure of Heiwadai, with reopening undetermined, excludes it. Miharadai's announced February–May 2027 closure is a future change and does not remove it from the current stock [4]. Counting venues once avoids adding lanes or separate basins. With the unchanged supplied population of 9,983,365, the existing pool anchor 1 gives $[0.04769460951, 1]$. Other wards and

providers remain unknown. Routine maintenance days and restricted sessions mean that this stock count does not imply simultaneous opening or individual adolescent access.

Direct further searches toward decisions

A separate acquisition diagnostic holds the pre-update V121 evidence fixed. It distinguishes reducing the possible-champion set from changing the conservative recommendation. An exact, correctly scoped Dublin pupil–teacher ratio below 17.58238 would change the fixed-weight recommendation to Dublin; this is a conditional threshold, not an observed ratio. Zurich remains the fixed-weight recommendation for an exact ratio below 175.33693 and remains the equal-weight choice throughout the admissible ratio range. Interval-valued teacher evidence requires the full two-bound decision calculation.

A matched teacher denominator remains the key requirement. The inspected Zurich series covers canton-level publicly funded schools [5]; Dublin's official report says school-level FTE are not published [6]. These sources do not establish a citywide, matching teacher denominator. Their missing cells remain unknown; a source's existence alone does not make its population compatible.

Controlled verification and manuscript scope

Independent weighted norms and explicit competitor exclusion reproduce twenty static comparisons and twenty-four weight-simulation settings, each with 10,000 draws. The original random streams are retained, and the pre-update arm matches saved V121 results. Withdrawing both supplements restores every V121 utility bound. Weight-simulation frequencies remain conditional on their chosen distributions, rather than actual winning probabilities.

This companion supplies new data-collection and baseline evidence. The V123 complete review paper and V122 preference, future and method results retain their explicitly frozen V121 inputs. Their numerical claims have not been silently relabelled V124; integrating the new evidence requires rerunning those extensions and updating the paper consistently. Team control and the complete AI-use record remain pending, so this companion is research manuscript material, not a final submission. OpenAI Codex assisted extraction checks, calculation and writing [7]; it does not provide missing empirical observations.

References

[1] Consorcio Regional de Transportes de Madrid. *M4 Estaciones*, record date 29 May 2026, accessed 9 November 2026. `https://services5.arcgis.com/UxADft6QPcvFyDU1/arcgis/rest/services/M4_Red/FeatureServer/0`.

[2] Instituto Nacional de Estadística. Transport statistics metadata, municipal classification, 2026. `https://www.ine.es/dynt3/metadatos/RespuestaDatos.html?oper=297`.

[3] Nerima City. Municipal swimming facilities and linked venue pages, accessed 9 November 2026. `https://www.city.nerima.tokyo.jp/shisetsu/koen/index.html`.

[4] Nerima City. Heiwadai closure notice, 27 October 2026; Miharadai planned-closure notice, 29 September 2026. https://www.city.nerima.tokyo.jp/kankomoyoshi/sports/oshirase/20261027.html; https://www.city.nerima.tokyo.jp/kankomoyoshi/sports/oshirase/20260929.html.

[5] Canton of Zurich. Development of teaching staff at middle and vocational schools, 2026. https://www.zh.ch/de/bildung/bildungssystem/zahlen-fakten/entwicklung-des-schulpersonals-an-mittelschulen-und-berufsfachschulen.html.

[6] Department of Education and Youth, Ireland. *Teacher Workforce Data Quality Report*, 2026, Sections 5.2 and 8.4. https://nationaldata.ie/media/NationalDataie/MethodologyDocumentation/Teacher_Workforce_Data_Quality_Report.pdf.

[7] OpenAI. Codex, research, calculation and writing assistance, November 2026; exact deployed model build not retained in draft metadata. https://openai.com/codex/.